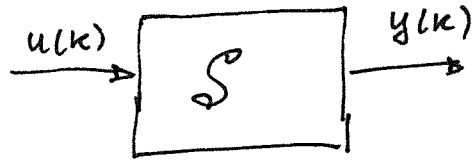


$\Rightarrow$ NON LINEAR ARX (NARX)

(F122)

HP:



$$y(k) = \gamma[\varphi(k)] + u(k)$$

γ = unknown nonlinear function

$$\varphi(k) = [y(k-1) \dots y(k-n) \ u(k) \dots u(k-m)]^T$$

Can be interpreted
as a nonlinear difference
equation.

Ideally: best model for S would be $\hat{y}(k) = \gamma[\varphi(k)]$

But γ is unknown $\Rightarrow$ Instead of searching $\gamma \in \Gamma$
(Γ could be $C^0, C^1, C^2, \dots$)

we use an approximate structure for the
function

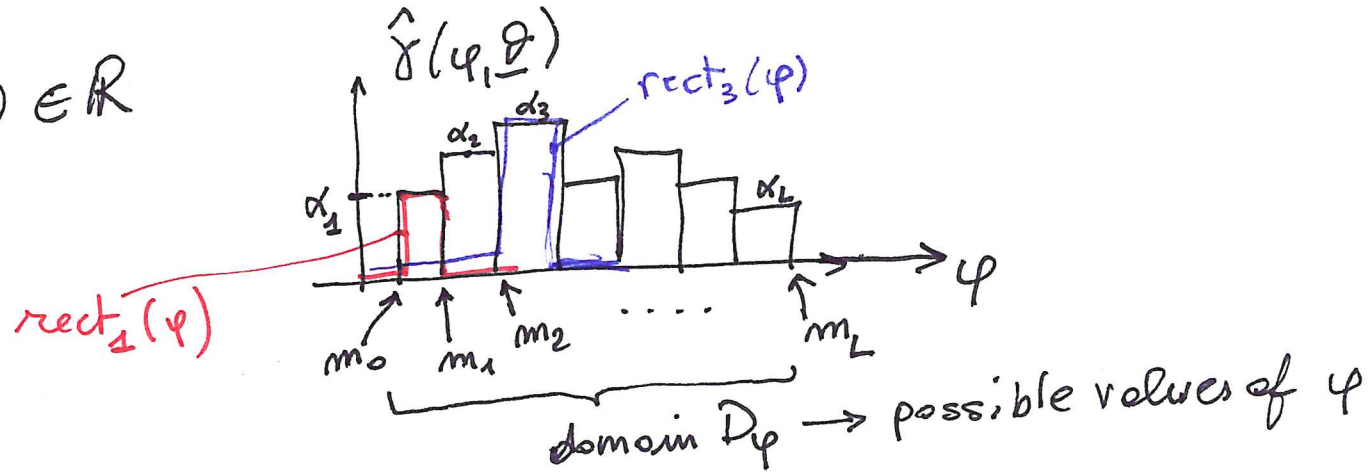
$$\gamma(\varphi(k)) \rightsquigarrow \hat{\gamma}[\varphi(k), \underline{\theta}]$$

where $\hat{\gamma}$ is a function of $\varphi(k)$, ~~para~~ with some

pre-determined structure depending on a set of parameters. (I 123)

ES .

$$\varphi(k) \in \mathbb{R}$$

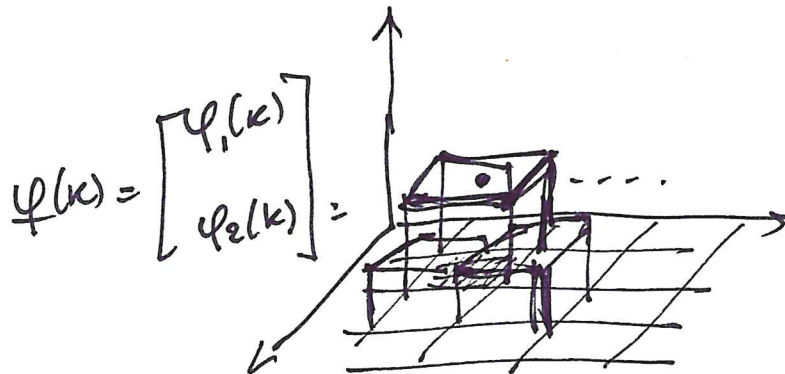


If $m_0, \dots, m_L$ are set a-priori $\Rightarrow$ parameters are
 $\underline{\theta} = [\alpha_1 \dots \alpha_L]^T$

We can easily extend this representation to the multi-dim.

core

ES

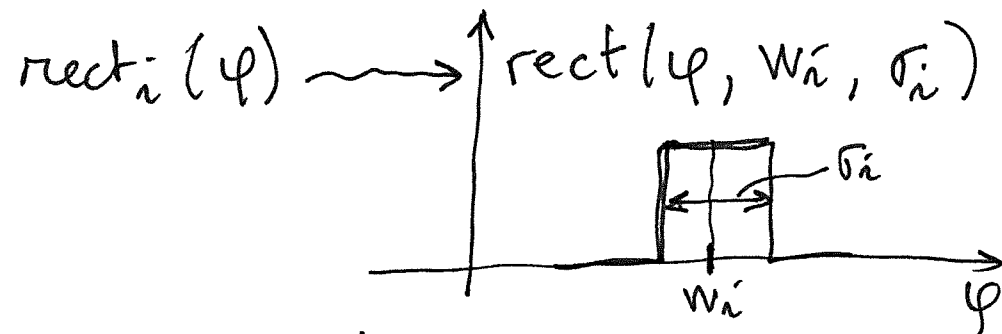


Referring to the scalar case we could write $(\varphi(x) \rightsquigarrow \varphi)$ (3.124)

$$\gamma(\varphi, \underline{\theta}) = \sum_{\hat{n}=1}^L \alpha_{\hat{n}} \text{rect}_{\hat{n}}(\varphi)$$

$$\text{where } \text{rect}_{\hat{n}}(\varphi) = \begin{cases} 1 & \text{if } \varphi \in [m_{\hat{n}-1}, m_{\hat{n}}) \\ 0 & \text{else} \end{cases}$$

Of course also $m_0 \dots m_L$ can be considered as parameters:

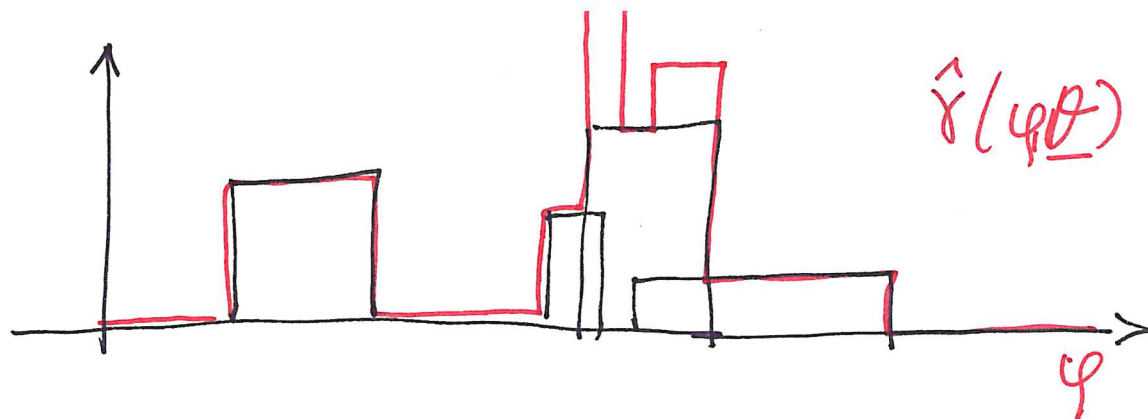


$$\Rightarrow \hat{\gamma}(\varphi, \underline{\theta}) = \sum_{\hat{n}=1}^L \alpha_{\hat{n}} \text{rect}(\varphi, \underbrace{w_{\hat{n}}, \sigma_{\hat{n}}}_{\underline{\tau}_{\hat{n}} = [w_{\hat{n}}, \sigma_{\hat{n}}]}) = \sum_{\hat{n}=1}^L \alpha_{\hat{n}} \text{rect}(\varphi, \underline{\tau}_{\hat{n}})$$

$\underline{\tau}_{\hat{n}} \quad \hat{n}=1 \dots L$ internal parameters

~~rect~~ $(\varphi, \underline{\tau}_{\hat{n}})$ are "basis functions".

$\text{rect}(\cdot, \cdot)$ here $\underline{\theta} = [\alpha_1, w_1, \sigma_1, \dots, \alpha_L, w_L, \sigma_L]$



Of course, augmenting the number of param. augments our approx. capability.

Other possible representations: Sum of polynomials

$$\text{Scalar case } \hat{y}(\varphi, \underline{\theta}) = \sum_{i=0}^L \alpha_i \varphi^i =$$

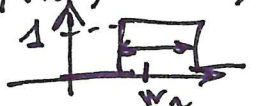
$$= \alpha_0 + \alpha_1 \varphi + \dots + \alpha_L \varphi^L$$

$$\underline{\theta} = [\alpha_0 \dots \alpha_L]^T$$

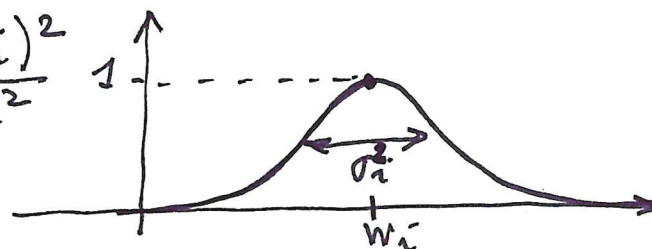
Instead of rectangles $\rightsquigarrow$ we can use bell-shaped functions.

scalar case

$\text{rect}(\varphi, w_i, \sigma_i)$

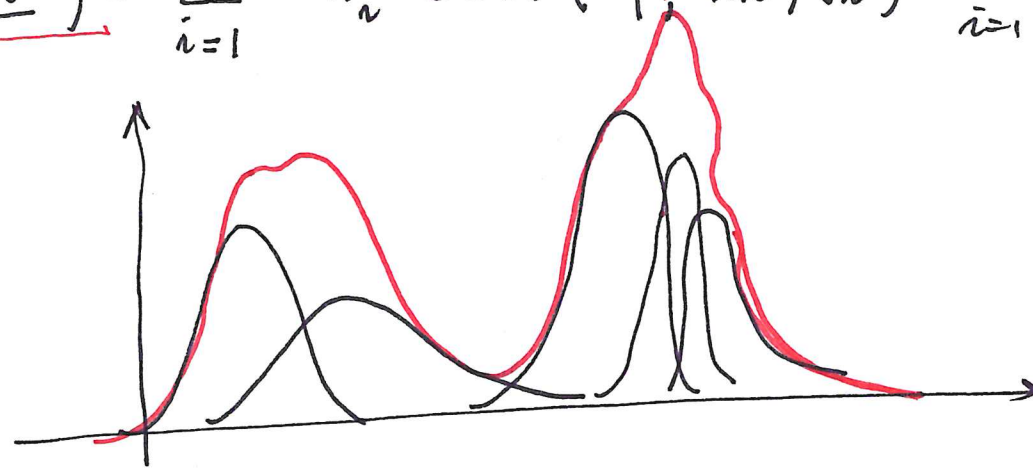


$$e^{-\frac{(\varphi - w_i)^2}{\sigma_i^2}}$$



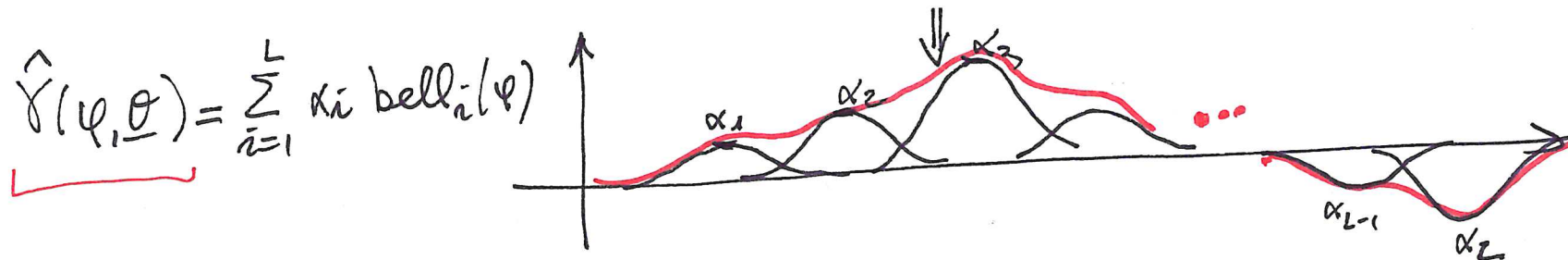
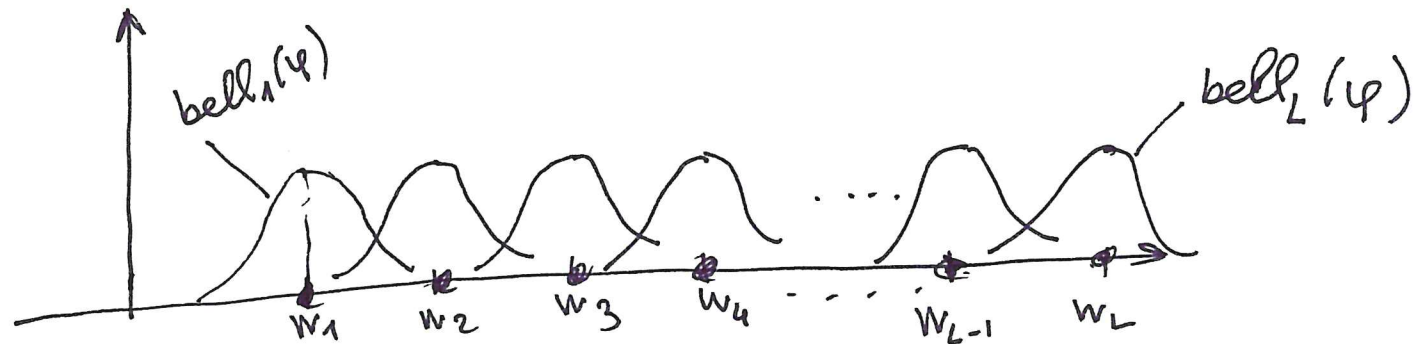
$$\Rightarrow \underline{\hat{\gamma}(\varphi, \underline{\theta})} = \sum_{i=1}^L \alpha_i \text{bell}(\varphi, w_i, \sigma_i) = \sum_{i=1}^L \alpha_i \text{bell}(\varphi, \underline{\tau_i})$$

$$\underline{\tau_i} = \begin{bmatrix} w_i \\ \sigma_i \end{bmatrix}$$



in the case $w_i, \sigma_i \quad i=1 \dots L$ are pre-set (a priori)

$$\Rightarrow \text{bell}(\varphi, w_i, \sigma_i) \longrightarrow \text{bell}_i(\varphi)$$

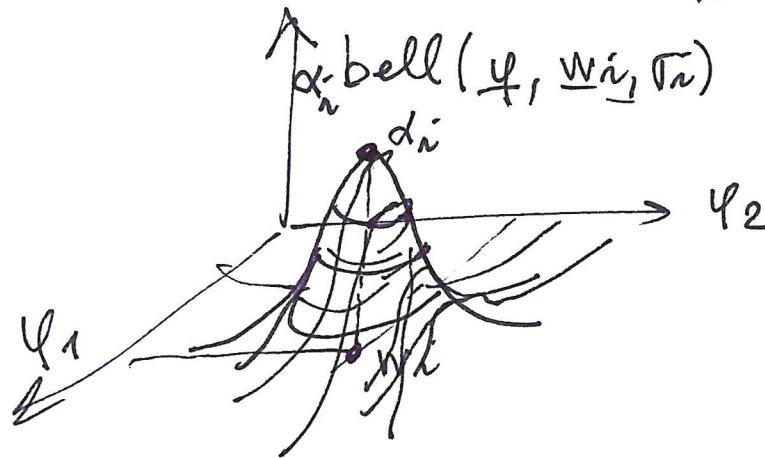


Sum of bells = Gaussian functions or more generally sum of functions with radial symmetry: RBF = Radial Basis Functions. (I 127)

RBFs can be easily extended to multi-dim. case.

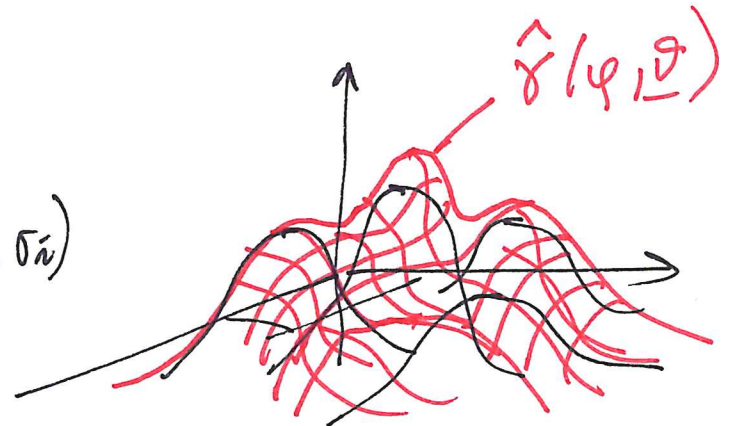
$$\begin{aligned}\hat{y}(\underline{y}, \underline{\theta}) &= \sum_{\bar{n}=1}^L \alpha_{\bar{n}} \text{bell}(\underline{y}, \underline{w}_{\bar{n}}, \sigma_{\bar{n}}) = \\ &= \sum_{\bar{n}=1}^L \alpha_{\bar{n}} \underbrace{e^{-\frac{\|\underline{y} - \underline{w}_{\bar{n}}\|^2}{\sigma_{\bar{n}}^2}}}_{\text{bell}(\underline{y}, \underline{w}_{\bar{n}}, \sigma_{\bar{n}})}\end{aligned}$$

dim $\underline{y} = 2$



$$\hat{y}(\underline{y}, \underline{\theta}) = \sum_{\bar{n}=1}^L \alpha_{\bar{n}} \text{bell}(\underline{y}, \underline{w}_{\bar{n}}, \sigma_{\bar{n}})$$

here $\underline{\theta} = \{ \alpha_{\bar{n}}, \underline{w}_{\bar{n}}, \sigma_{\bar{n}} \dots \bar{n} = 1 \dots L \}$



$$\underline{\theta} = \text{col} \{ \alpha_i, \underline{w}_i, \sigma_i, \quad i = 1 \dots L \} = \begin{bmatrix} \alpha_1 \\ \underline{w}_1 \\ \sigma_1 \\ \alpha_2 \\ \underline{w}_2 \\ \sigma_2 \\ \vdots \end{bmatrix}$$

(I, 28)

~~7/8/~~ If internal parameters are pre-set (a-priori)

$$\hat{y}(\varphi, \underline{\theta}) = \sum_{i=1}^L \alpha_i \text{bell}_i(\varphi)$$

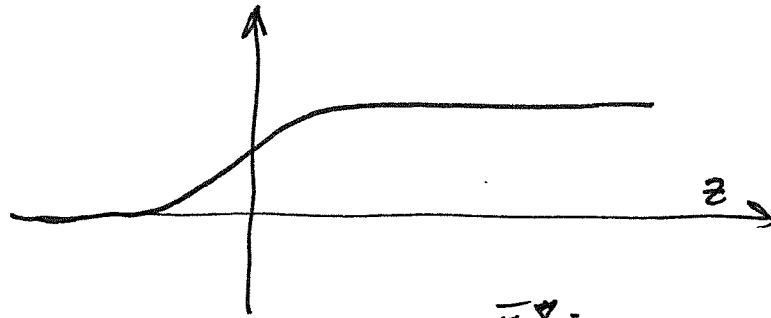
$$\text{bell}_i(\varphi) = e^{-\frac{\|\varphi - \underline{w}_i\|^2}{\sigma_i^2}}$$

here $\underline{\theta} = [\alpha_1 \dots \alpha_L]^T$

• FFNN (Feed Forward Neural Network) with one "hidden layer"

$$\hat{y}(\varphi, \underline{\theta}) = \sum_{i=1}^L \alpha_i \text{sigm}(\varphi^T \underline{w}_i + v_i)$$

$\text{sigm}(z) = \text{sigmoidal function}$ (^{non} decreasing with limits) (I 129)



EX: $\text{sigm}(z) = \frac{1}{1 + e^{-z}}$

General representation:

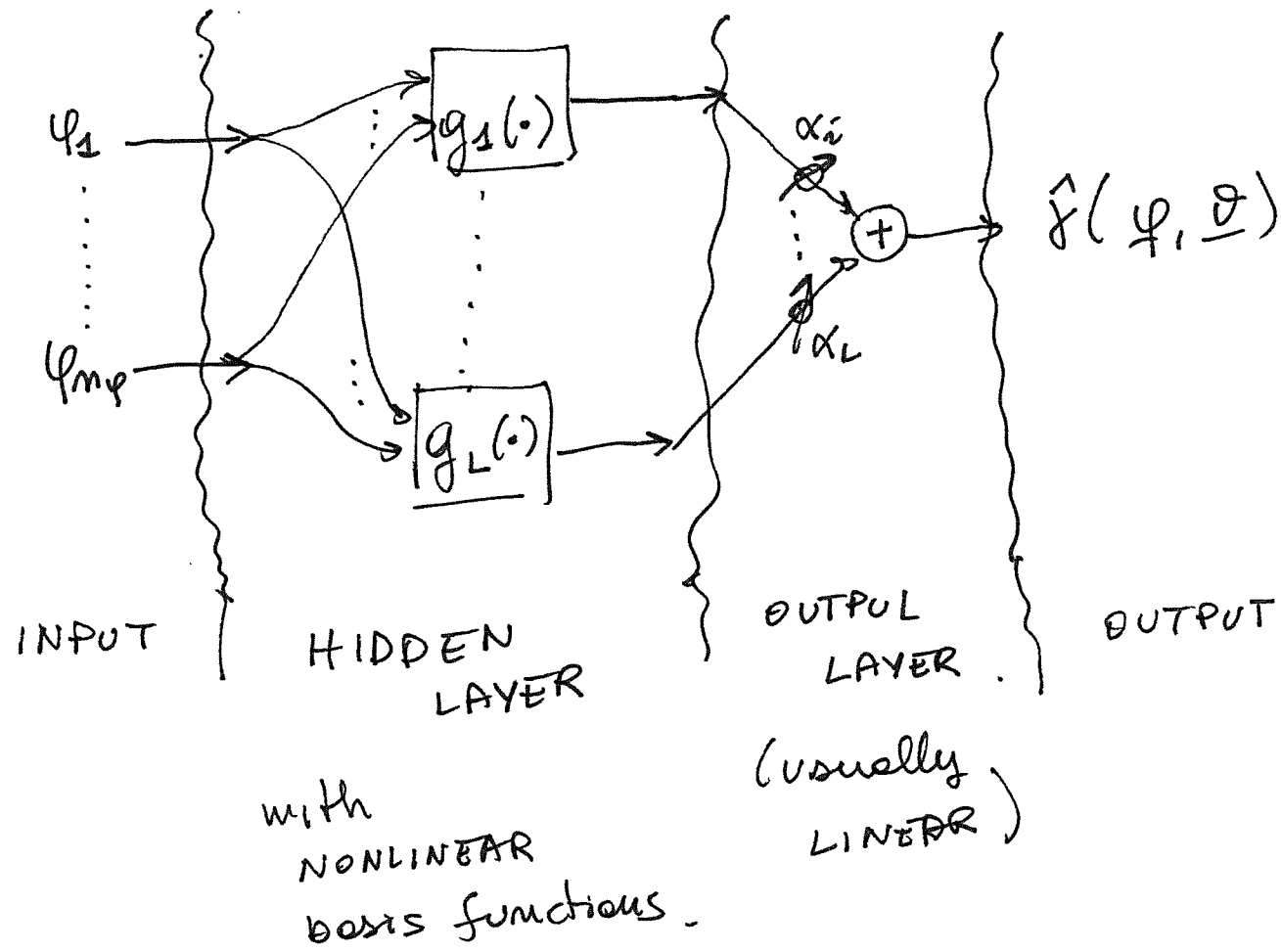
O. H. L.
(one - Hidden Layer)
functions

$$\hat{f}(\varphi, \underline{\theta}) = \begin{cases} \sum_{i=1}^L \alpha_i g_i(\varphi) & \text{(A)} \\ \sum_{i=1}^L \alpha_i g(\varphi, \underline{\tau}_i) & \text{(B)} \end{cases}$$

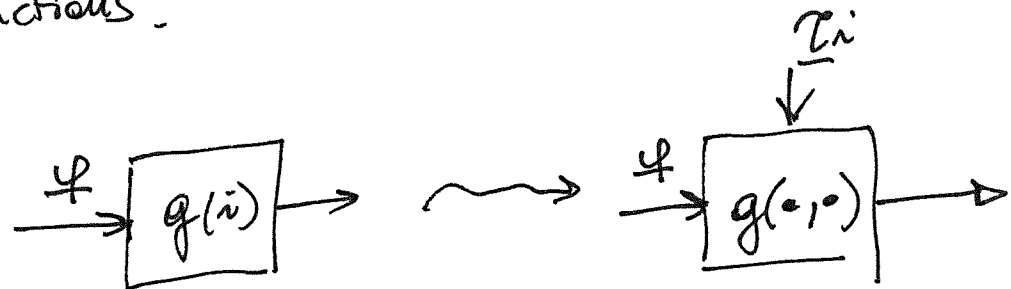
$g_i(\cdot)$ or $g(\cdot, \cdot)$ are called basis functions.

$\alpha_i = \text{weights}$ $\underline{\tau}_i = \text{internal parameters.}$

(A)



Similarly in case (B)



Several more hidden layers can be put in series

$\Rightarrow$ MULTI-"HIDDEN LAYER" networks.
("NEURAL NETS")

We concentrate on the use of OHL functions ~~for~~ for identification.

Let assume we have chosen a particular approximate structure for prediction.

for an "estimated" ~~$\hat{y}(k)$~~ , param. vector $\underline{\hat{\theta}}$

$$\hat{y}(k | \underline{\hat{\theta}}) = \begin{cases} \sum_{i=1}^L \alpha_i g_i[\psi(k)] & \textcircled{A} \\ \sum_{i=1}^L \alpha_i g[\psi(k), \underline{\tau}_i] & \textcircled{B} \end{cases}$$

Ⓐ usually referred to the linear-in-the-parameters case

Ⓑ " " " " NON-lin. " " " "

in fact . Ⓐ: $\underline{\hat{\theta}} = [\alpha_1, \dots, \alpha_L]$

Ⓑ $\underline{\hat{\theta}} = \text{col}[\alpha_1, \underline{\tau}_1, \alpha_2, \dots, \alpha_L, \underline{\tau}_L]$

We will use $p = \dim \underline{\theta} \Rightarrow$

Ⓐ $p = L$

Ⓑ $p = L(1 + \dim \underline{\tau}_i)$

Pr ① (p=L) $\hat{y}(k|\hat{\theta}) = \sum_{\hat{n}=1}^p \hat{\alpha}_{\hat{n}} g_{\hat{n}}(\underline{\varphi}(k)) =$

$$= \hat{\alpha}_1 g_1(\underline{\varphi}(k)) + \dots + \hat{\alpha}_L g_L(\underline{\varphi}(k)) =$$

(is a linear comb.
of $\alpha_1 \dots \alpha_L$ weighted
by $g_1(\underline{\varphi}(k)) \dots g_L(\underline{\varphi}(k))$)

$$= [g_1(\underline{\varphi}(k)) \dots g_L(\underline{\varphi}(k))] \begin{bmatrix} \hat{\alpha}_1 \\ \vdots \\ \hat{\alpha}_L \end{bmatrix} =$$

$$= g(\underline{\varphi}(k))^T \hat{\underline{\theta}}$$

where $\hat{\underline{\theta}} \triangleq \begin{bmatrix} \hat{\alpha}_1 \\ \vdots \\ \hat{\alpha}_L \end{bmatrix}$

$$g[\underline{\varphi}(k)] \triangleq \begin{bmatrix} g_1[\underline{\varphi}(k)] \\ \vdots \\ g_L[\underline{\varphi}(k)] \end{bmatrix}$$

$g_1(\cdot) \dots g_L(\cdot)$
NONLIN. basis functions.

↑
Corresponds to the
mapping of the NONLIN
hidden layer

NOTE THAT in case (A) we have linearity with respect to both $\underline{\hat{\theta}}$ and $\cancel{\varphi(k)} g[\varphi(k)]$

(1133)

↑
NON linear

$\Rightarrow$ if $g_i(\cdot)$ are NONLINEAR, linearity is only w.r.t. $\underline{\hat{\theta}}$.

linear in $\underline{\hat{\theta}}$, but NONLIN in $\varphi(k)$

ARX is linear in both $\underline{\hat{\theta}}$ and $\varphi(k)$

ARX is a particular case of N-ARX where

$$g[\varphi(k)] \equiv \varphi(k)$$

$\varphi(k)$ is NOT nonlinearly modified

N-ARX (A)		ARX
$\hat{y}(k \underline{\hat{\theta}}) = g^T[\varphi(k)] \underline{\hat{\theta}}$		$\hat{y}(k \underline{\hat{\theta}}) = \varphi^T(k) \underline{\hat{\theta}}$

$\Rightarrow$ Structure is similar to ARX but $\varphi(k) \rightarrow g[\varphi(k)]$

$$\Rightarrow \hat{\underline{\theta}}^* = \arg \min_{\hat{\underline{\theta}}} J_N(\hat{\underline{\theta}}) = \arg \min_{\hat{\underline{\theta}}} \frac{1}{N} \sum_{k=1}^N [y(k) - \hat{y}(k|\hat{\underline{\theta}})]^2 \quad (\text{I134})$$

for some DATA SET $\{u(k), y(k), k=1 \dots N\}$
 and a structure $\hat{y}(k|\hat{\underline{\theta}}) = \hat{\gamma}[\psi(k), \hat{\underline{\theta}}] =$
 $= \sum_{i=1}^L \hat{\alpha}_i g_i(\psi(k)) =$
 $= \underline{g}[\psi(k)]^T \hat{\underline{\theta}}$

mimicking the ARX solution

$$\psi(k) \rightsquigarrow \underline{g}[\psi(k)]$$

ARX: $\hat{\underline{\theta}}^* = (\Phi_N \Phi_N^T)^{-1} \Phi_N y_N$ where $\Phi_N = [\psi(1) | \dots | \psi(N)]$

NARX: $\boxed{\hat{\underline{\theta}}^* = (G_N G_N^T)^{-1} G_N y_N}$ where $G_N = [\underline{g}[\psi(1)] | \dots | \underline{g}[\psi(N)]]$
 $y_N = [y(1) \dots y(N)]^T$